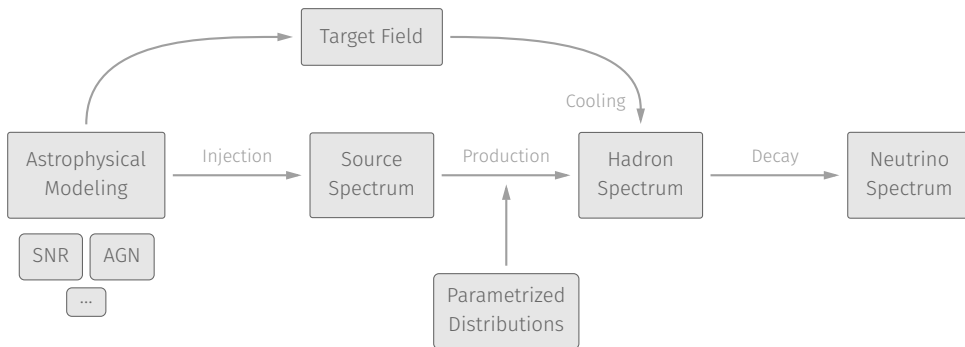

Neutrinos via Charm Decays in Astrophysical Sources

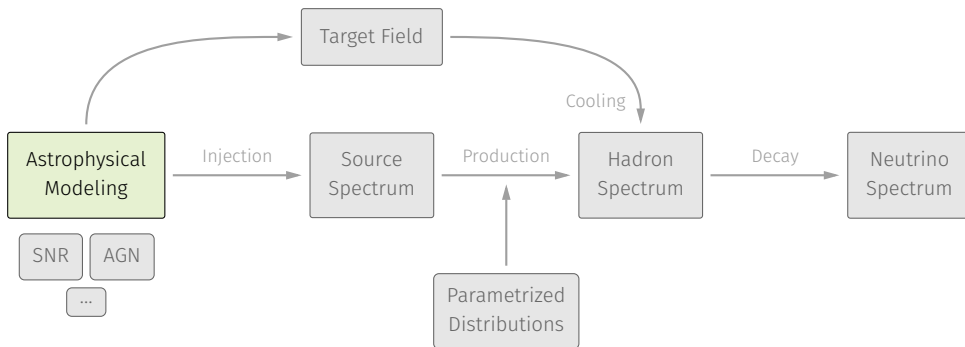
TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

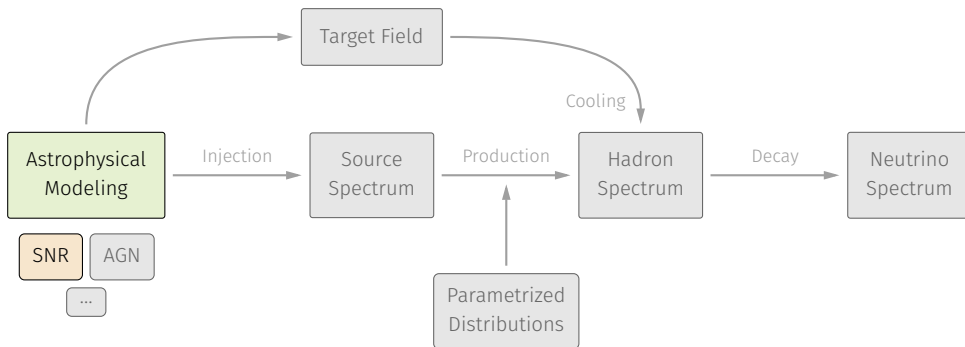
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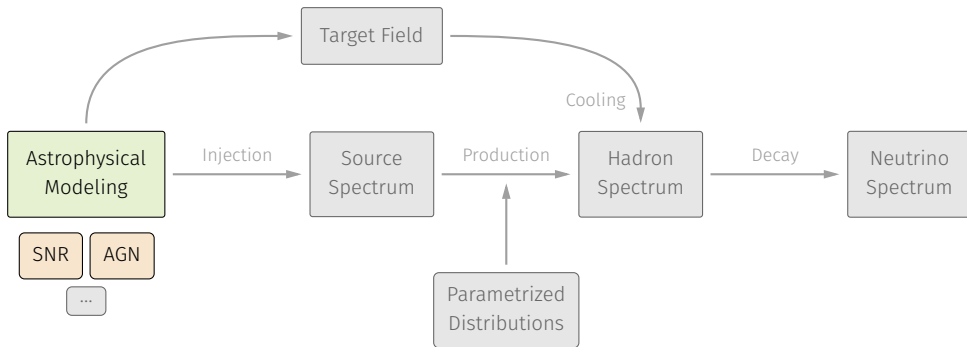
September 3, 2024

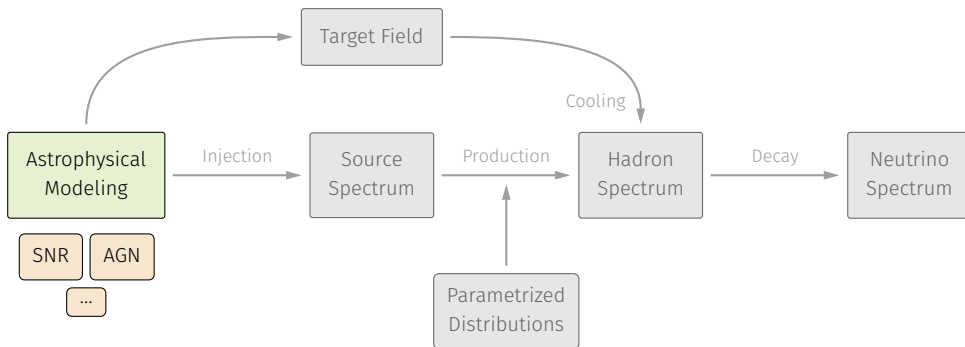
Technical University of Dortmund & Ruhr University Bochum

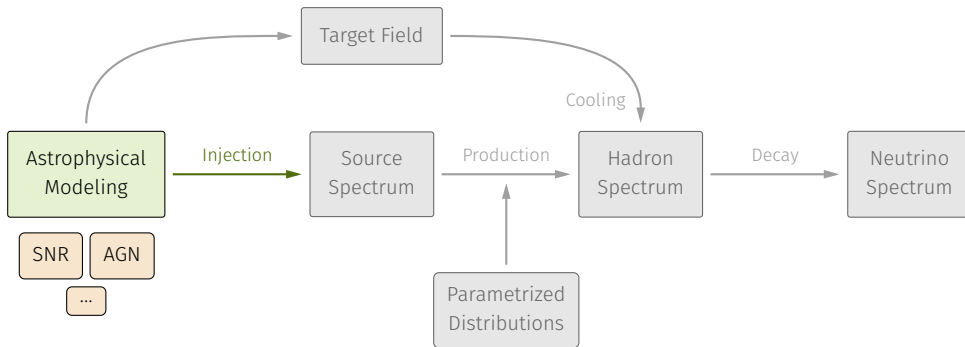


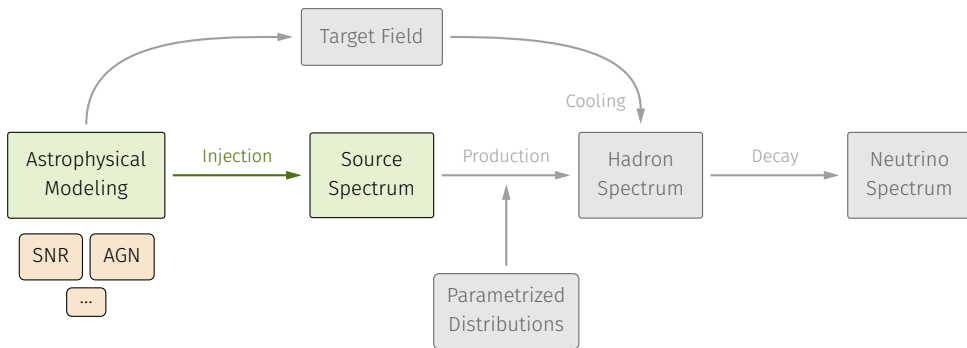


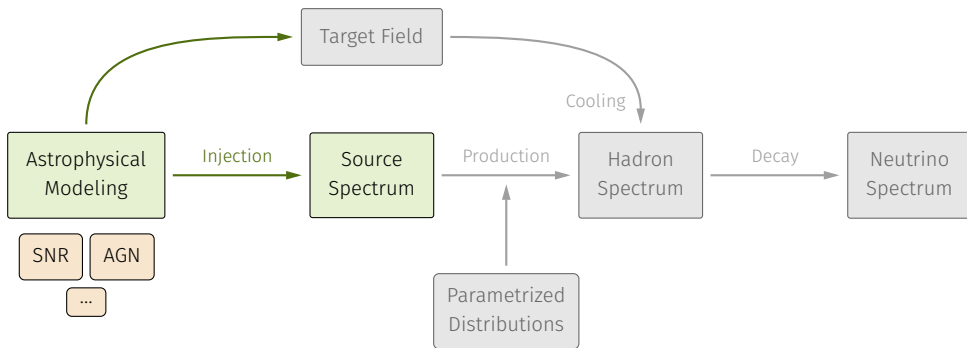


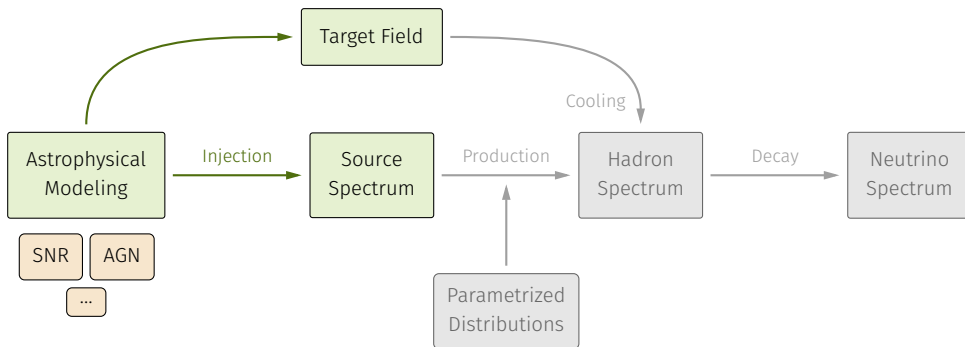


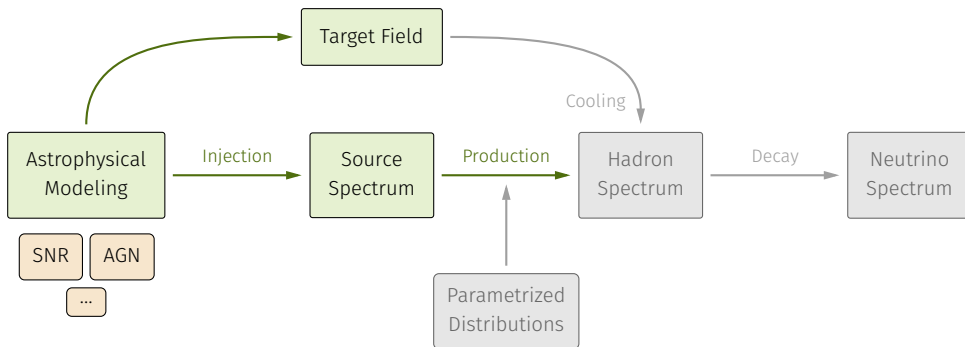


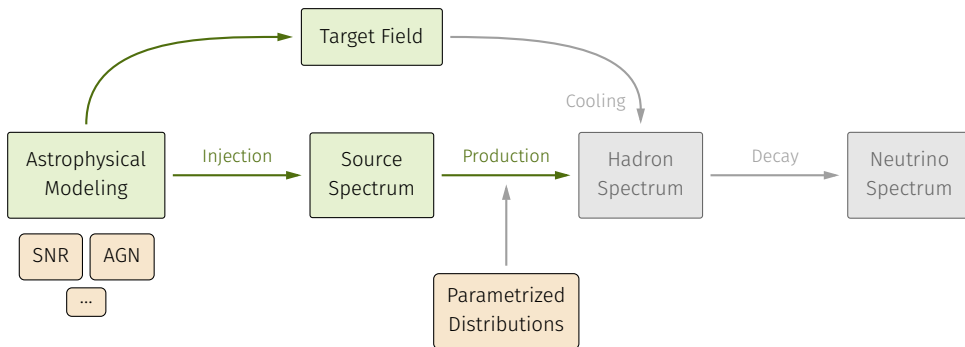


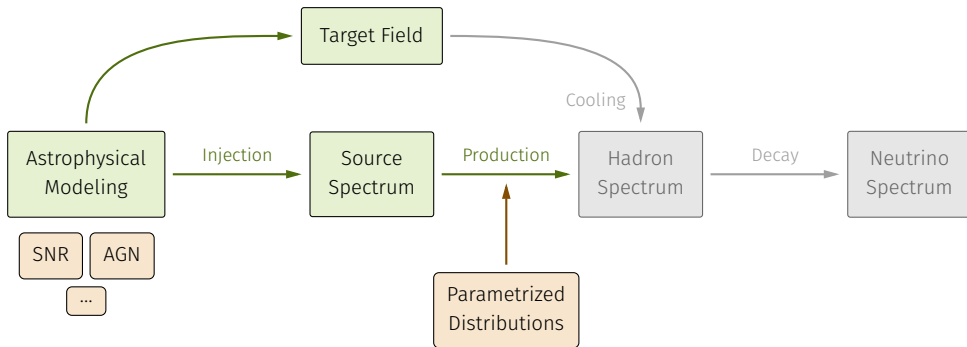


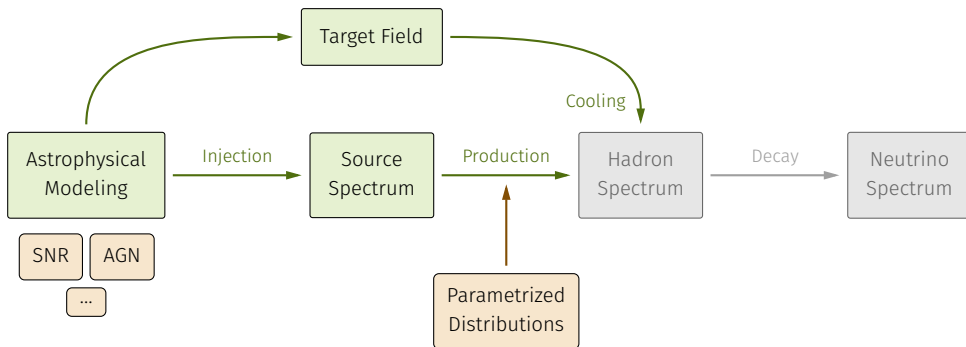


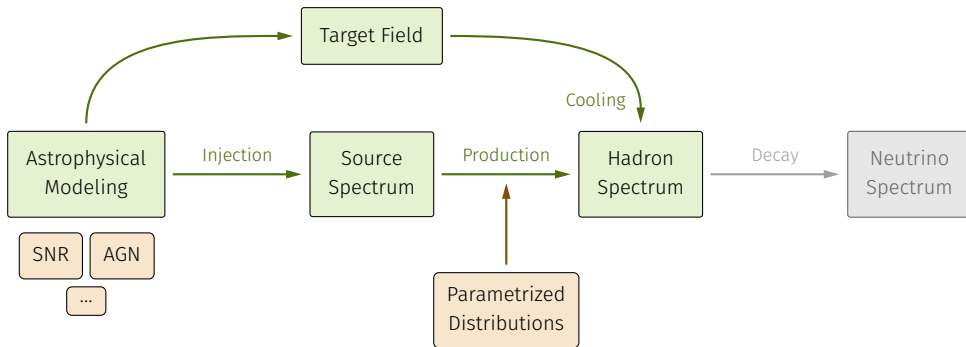


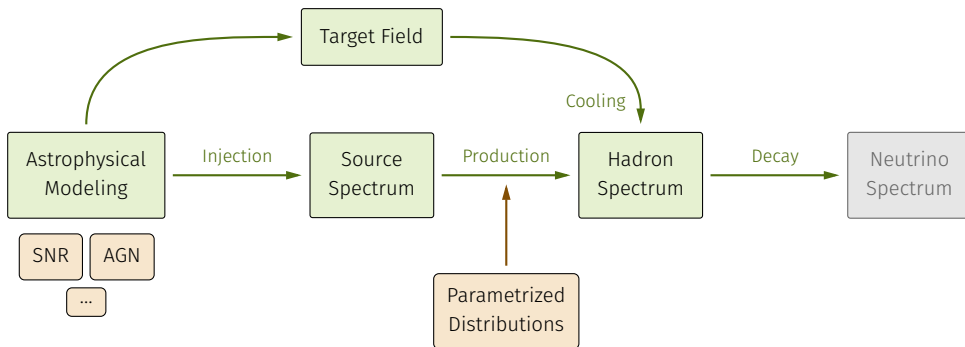


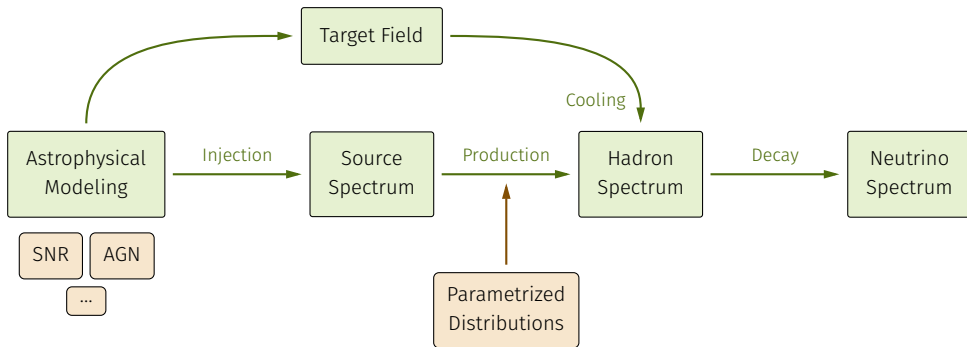


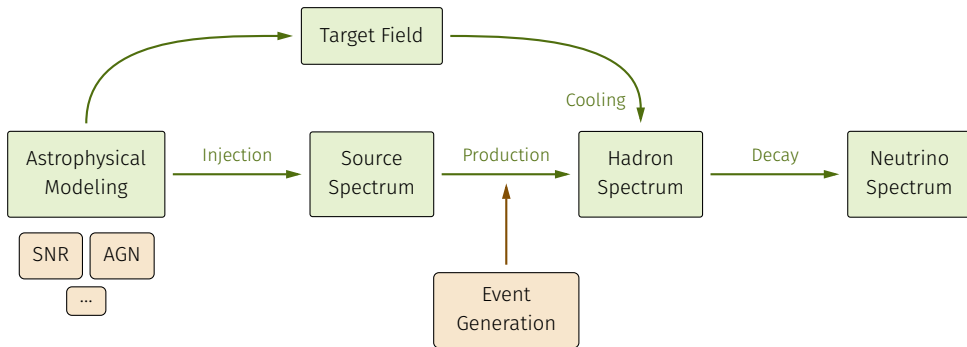




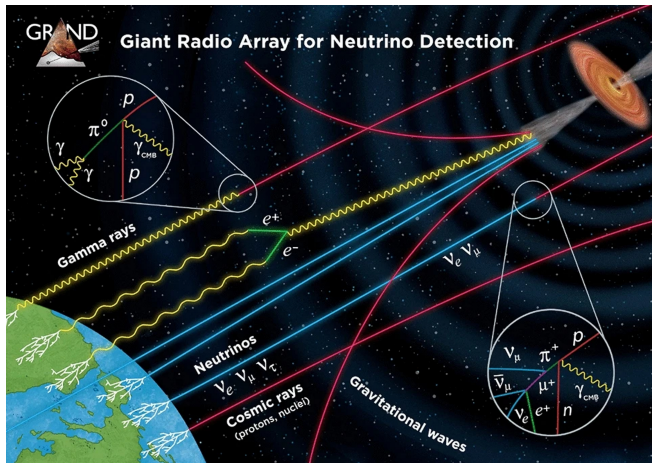






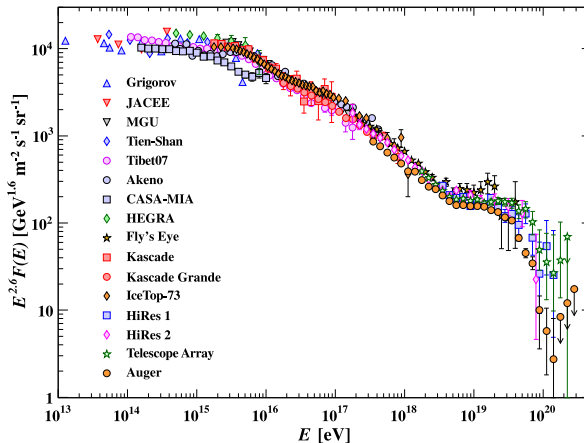


Multimessenger Astrophysics



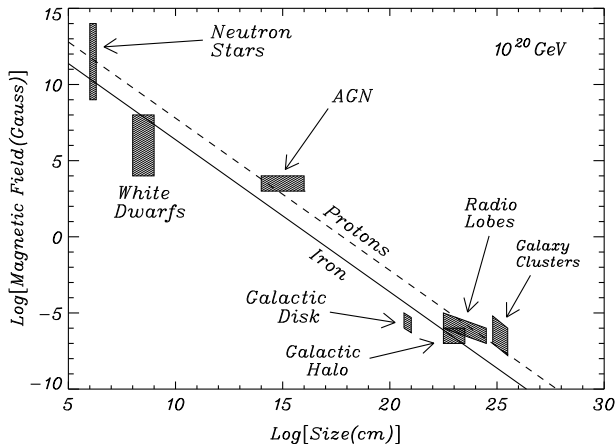
X. L. Qian et al.
Nucl. Sci. Tech. (2021)

Cosmic Ray Spectrum & Cutoff



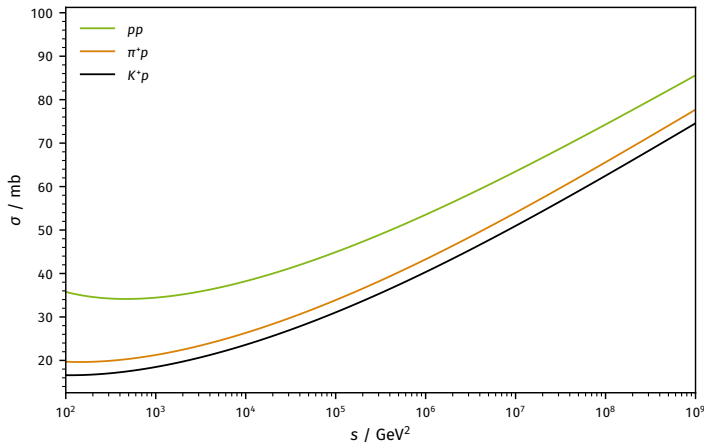
K. A. Olive et al.
Chin. Phys. C (2014)

Possible High Energy Cosmic Ray Origins



A. V. Olinto
AIP Conf. Proc. (2000)

Hadron Inelastic Scattering Cross Section

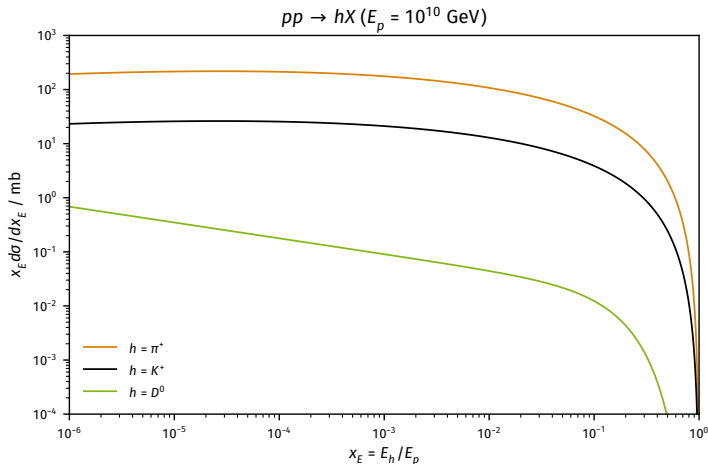


Combined Parametrizations

V. I. Belousov et al.
Phys. Atom. Nuclei (2016)

D. A. Fagundes, M. J. Menon
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Weighted Energy Differential Cross Section

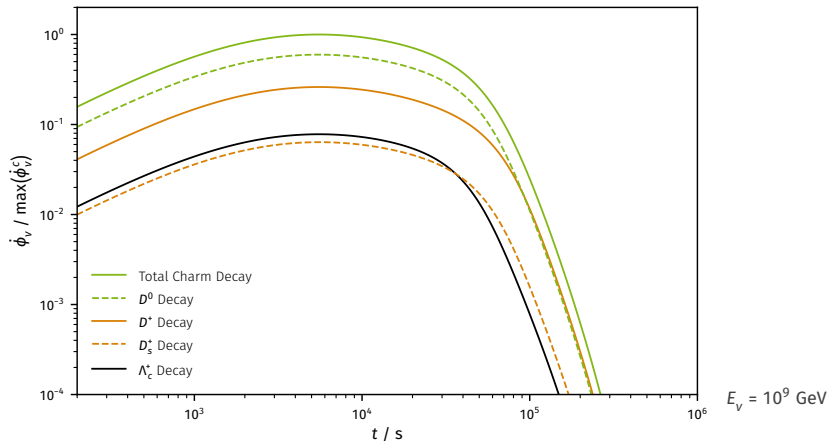


Derived Parametrizations

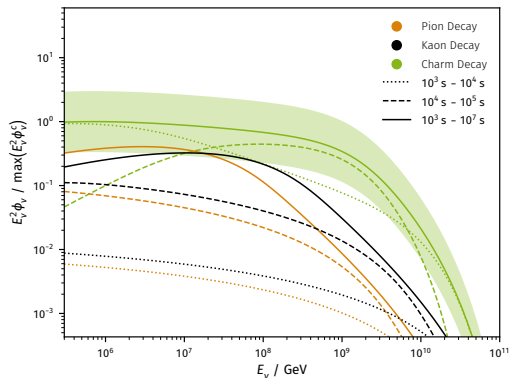
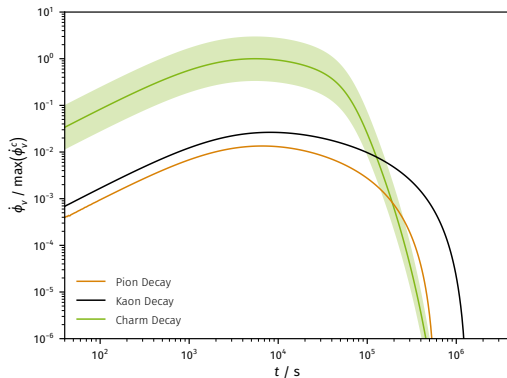
S. R. Kelner et al.
Phys. Rev. D (2006)

V. P. Gonçalves et al.
JHEP (2007)

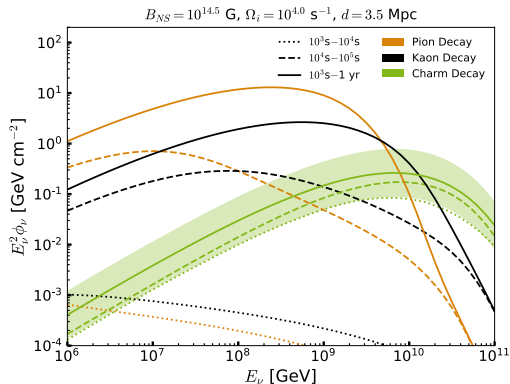
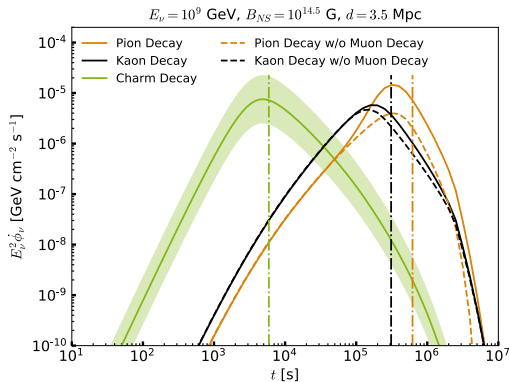
Normalized Magnetar Charm Contributions



Normalized Magnetar Neutrino Flux & Fluence

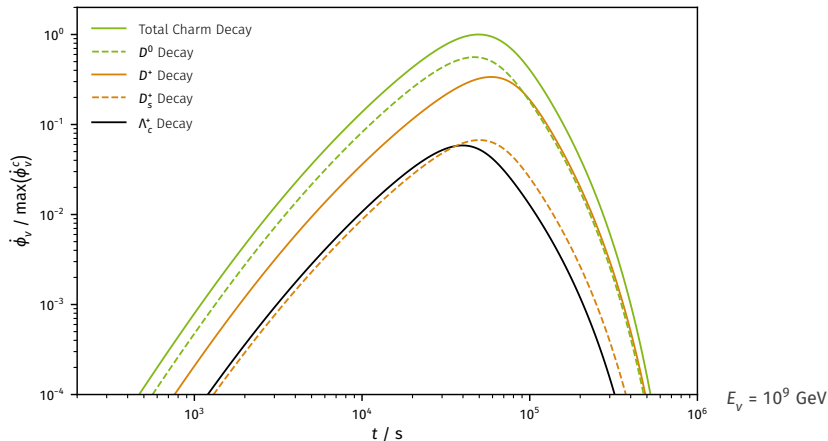


Comparison: Published Estimated Magnetar Neutrinos

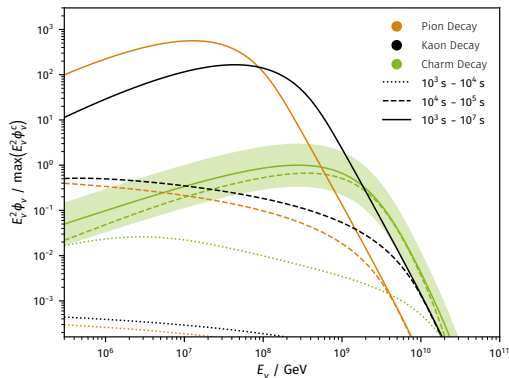
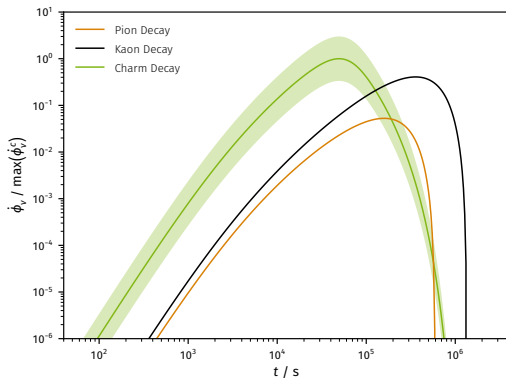


J. A. Carpio et al. *Phys. Rev. D* (2020)

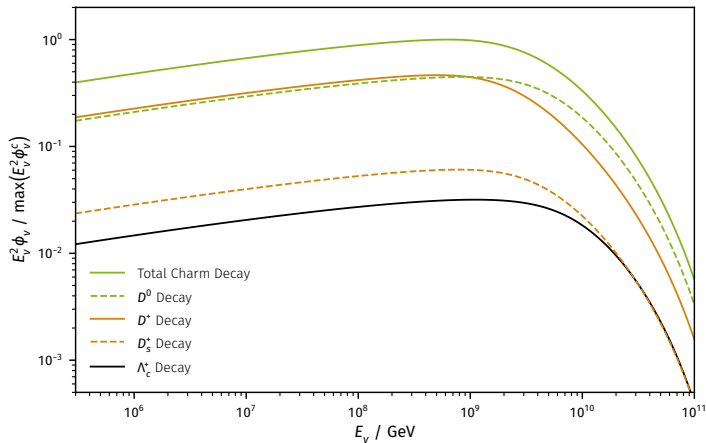
Modified Magnetar Charm Contributions



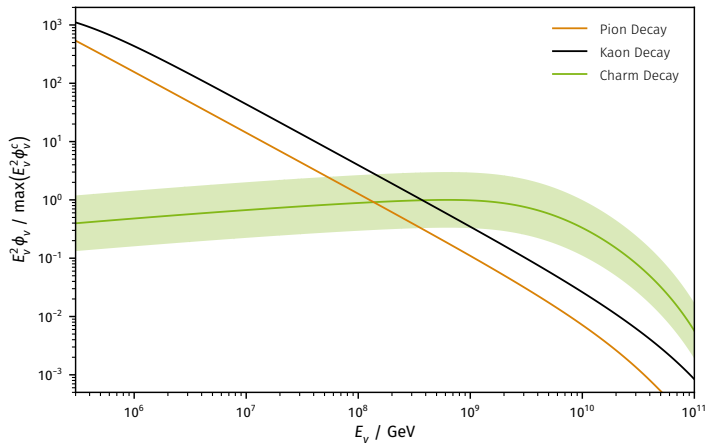
Modified Magnetar Neutrino Flux & Fluence



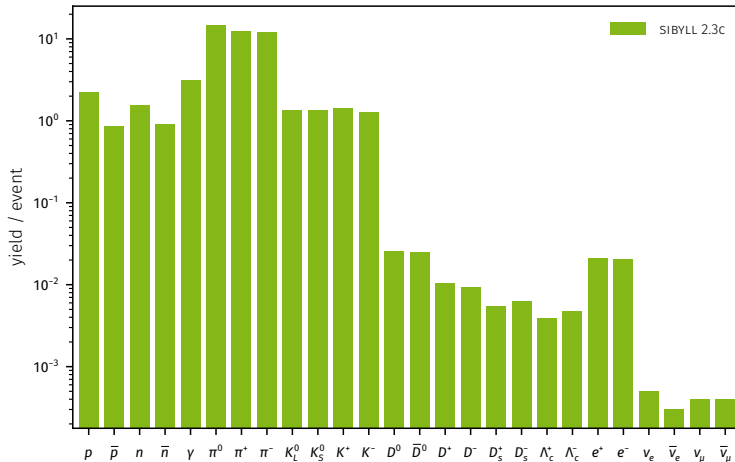
Normalized AGN Charm Contributions



Normalized AGN Neutrino Fluence

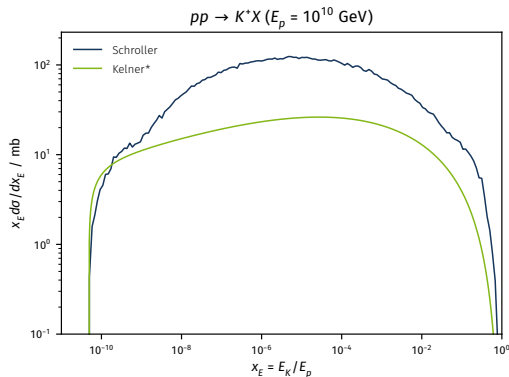
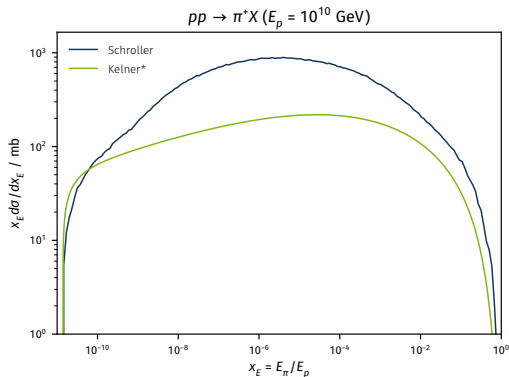


Monte Carlo Event Generation



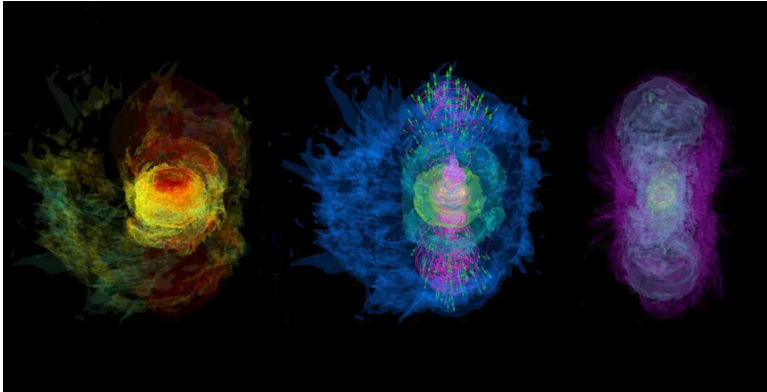
Data Courtesy
M. Schroller

Comparison: Differential Cross Section



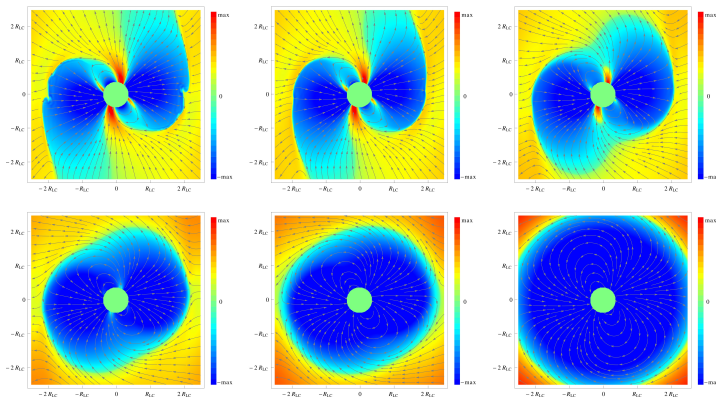
Derived Parametrization S. R. Kelner et al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schroll

Simulated Merger Magnetar Formation



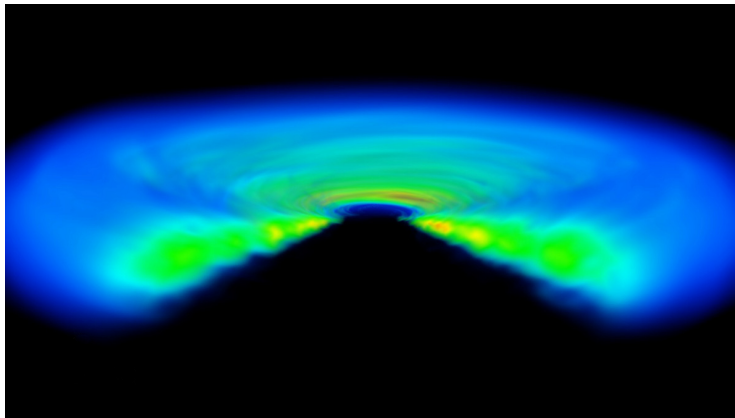
K. Hayashi *MPI Grav. Phys.* (2024)

Simulated Pulsar Magnetosphere



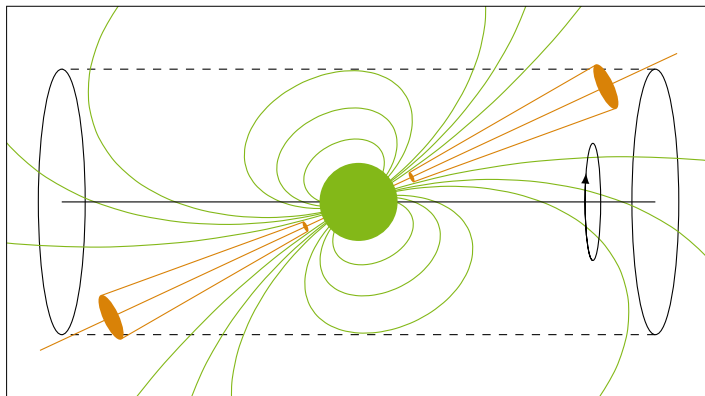
J. Li et al. *ApJ* (2012)

Simulated Accretion Disk Logarithmic Density



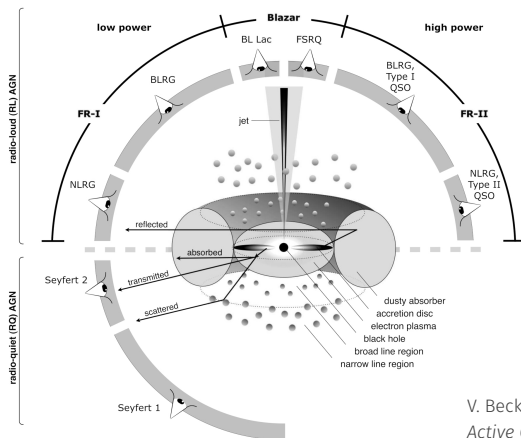
V. V. Zhuravlev et al. *ApJ* (2014)

Pulsar



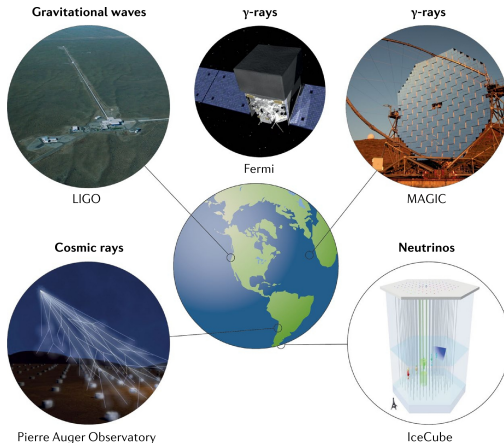
A. Knierim
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Unified AGN Classification Model



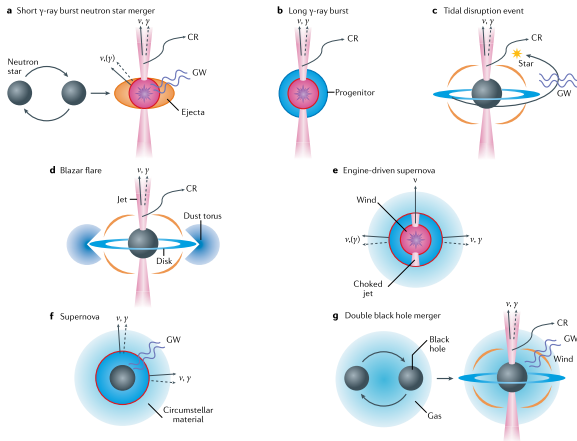
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

Detectors



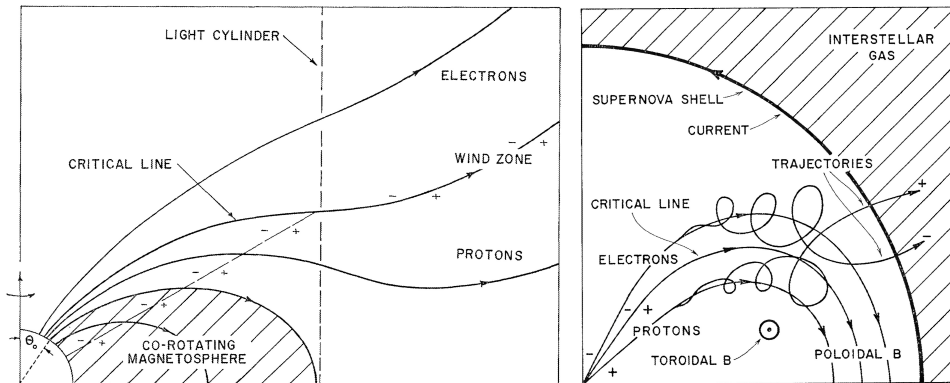
P. Mészáros et al.
Nat. Rev. Phys. (2019)

Sources



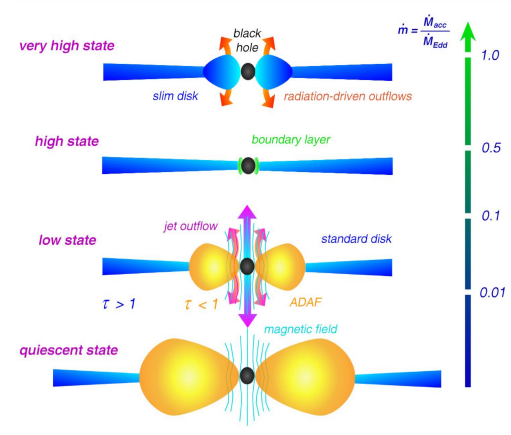
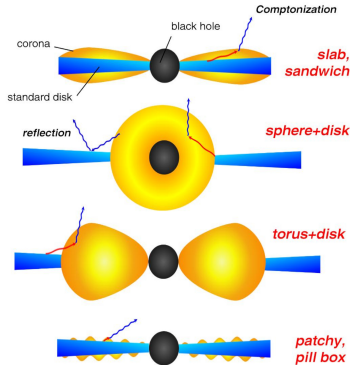
P. Mészáros et al.
Nat. Rev. Phys. (2019)

Pulsar Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)

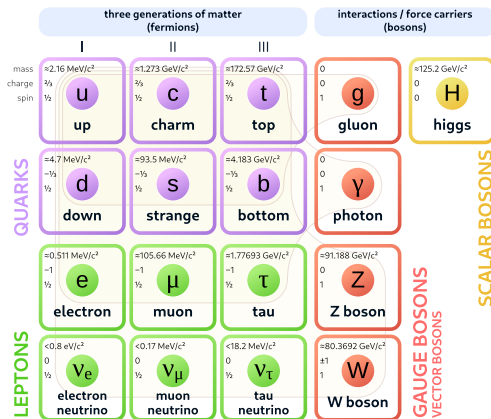
AGN Accretion Disk & Corona



A. Müller University Heidelberg (2004)

Elementary Particles

Standard Model of Elementary Particles



Standard Model
Wikimedia Commons (2019)